

Chapter 9

The Government and Fiscal Policy

Lecturer: Mihyun You

Department of Economics
Georgetown University

Summer 2024



GEORGETOWN UNIVERSITY

Contents

- 1 Government in the Economy
- 2 Government Budget and Government Deficit



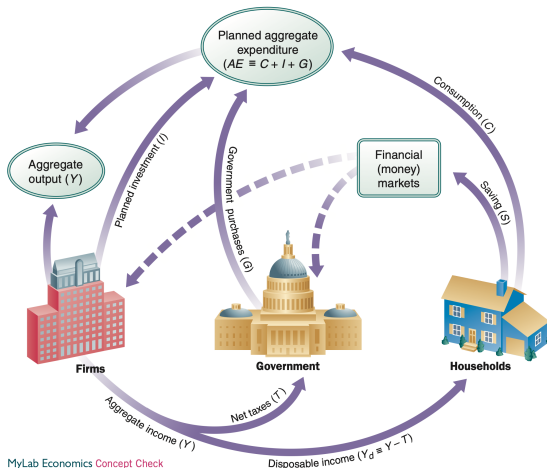
GEORGETOWN UNIVERSITY

Role of the Government

- Debate over whether the government should do in managing the macroeconomy
 - Classical: government spending is unnecessary, even destabilizing and harmful to economy.
 - Keynesian: economy without government intervention is likely to fluctuate too much, so the government should take actions to smooth the business cycles.
- Fiscal policy
 - government purchases of goods and services
 - taxing
 - transfer payments
- Monetary policy
 - interest rate policy controlled by the central bank

Adding Government Side to the Keynesian Model

- Circular flow of income with the government side



MyLab Economics Concept Check

Adding Taxes to the Consumption Function

- Note that government is not a producer of goods and services. Rather, it is another agent in the expenditure part.

$$AE \equiv C + I + G.$$

- Now, households, firms, and the government should split the produced total output Y , which means the consumption function should be adjusted.

$$\begin{aligned} C &= C(Y) \\ &= a + bY. \end{aligned}$$

$$\begin{aligned} \rightarrow C &= a + b \underbrace{(Y - T)}_{\text{disposable income}} \\ &= a + bY_d. \end{aligned}$$

The Determination of Equilibrium Output (Income)

- Imposing the equilibrium condition,

$$\begin{aligned} Y &= AE \\ &= C + I + G \\ &= a + b(Y - T) + I + G \end{aligned}$$

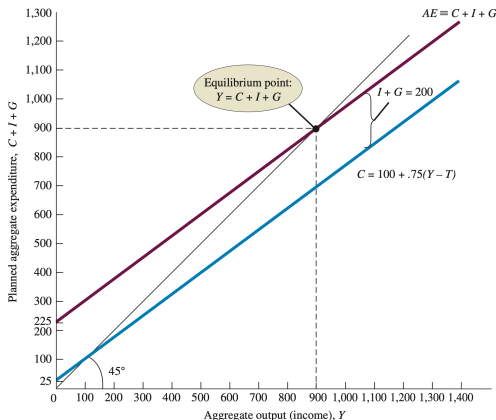
$$\rightarrow Y^* = \frac{a + I + G - bT}{1 - b}.$$

- Multiplier effects:
 - government spending multiplier
 - tax multiplier
 - balanced-budget multiplier

Graphic Analysis

- Numerical example:

$$a = 100, b = 0.75, I = 100, G = 100, T = 100$$



The Saving/Investment Approach to Equilibrium

- Recall the income/expenditure identity:

$$Y \equiv C + S + T$$

- In equilibrium,

$$Y = C + I + G$$

$$\rightarrow C + S + T = C + I + G$$

$$\rightarrow S + T = I + G.$$

- Note that equilibrium does NOT require that $G = T$ or that $S = I$.

Income Taxes

- The size of the multiplier is reduced when taxes depend on income.

$$T = T_0 + tY.$$

- Consumption function is adjusted to

$$\begin{aligned}C &= a + b(Y - T) \\&= a + b(Y - T_0 - tY) \\&= (a - bT_0) + b(1 - t)Y.\end{aligned}$$

- Finding the equilibrium output,

$$\begin{aligned}Y &= C + I + G \\&= (a - bT_0) + b(1 - t)Y + I + G \\ \rightarrow Y^* &= \frac{(a - bT_0) + I + G}{1 - b(1 - t)}.\end{aligned}$$

The Economy's Influence on the Government Budget

- Government budget deficit: $G - T$
- Automatic stabilizers and destabilizers
 - The automatic changes in government revenues and expenditures depending on the state of the economy
 - In recessions, taxes fall and expenditures rises, vice versa in expansions.
 - Inflation can be considered to be an automatic destabilizer.
- Full-employment budget: benchmark for evaluating fiscal policy.

Government Deficits

- Government deficits can be decomposed into two components.
- Structural deficit: the deficit that remains at full employment.
- Cyclical deficit: the deficit that occurs because of a downturn in the business cycle.